

Products

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
ProducerId	int	Yes	Exists in Producers	FK
CategoryId	int	Yes	Exists in Categories	FK
Name	varchar	Yes	Max 255 chars	
Description	text	No		
Price	decimal	Yes	≥ 0	
StockQuantity	int	Yes	≥ 0	
TraceabilityInfo	text	No		
IsAvailable	bool	Yes	true/false	
CreatedAt	datetime	Yes	Valid date	

Basket

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
UserId	string	Yes	Not Null	FK
CreatedAt	datetime	Yes	Valid date	

BasketItems

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
BasketId	int	Yes	Exists in Baskets	FK
ProductId	int	Yes	Exists in Products	FK
Quantity	int	Yes	> 0	

Favourites

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
UserId	string	Yes	Not Null	FK
ProductId	int	Yes	Exists in Products	FK
CreatedAt	datetime	Yes	Valid date	

Orders

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
UserId	string	Yes	Not Null	FK
OrderDate	datetime	Yes	Valid date	
OrderStatus	varchar	Yes	Predefined values	
TotalAmount	decimal	Yes	≥ 0	

OrderItems

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
OrderId	int	Yes	Exists in Orders	FK
ProductId	int	Yes	Exists in Products	FK
Quantity	int	Yes	> 0	
UnitPrice	decimal	Yes	≥ 0	

Deliveries

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
OrderId	int	Yes	Exists in Orders	FK
DeliveryType	varchar	Yes	Standard / Express	
DeliveryAddress	varchar	Yes	Max 255 chars	
ScheduledDate	datetime	Yes	Valid future date	
Status	varchar	Yes	Predefined values	

OrderTracking

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
OrderId	int	Yes	Exists in Orders	FK
Status	varchar	Yes		
UpdatedAt	datetime	Yes	Valid date	
Notes	text	No		

LoyaltyAccounts

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
UserId	string	Yes	Not Null	FK
Points	int	Yes	≥ 0	
Tier	varchar	Yes	Bronze/Silver/Gold	

LoyaltyTransactions

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
LoyaltyAccountId	int	Yes	Exists in LoyaltyAccounts	FK
PointsChange	int	Yes	Positive or negative	
Reason	varchar	Yes		
CreatedAt	datetime	Yes	Valid date	

Offers

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
Title	varchar	Yes	Max 255 chars	
Description	text	No		
DiscountPercent	decimal	Yes	0–100	
StartDate	datetime	Yes	Valid date	
EndDate	datetime	Yes	> StartDate	

ProductOffers

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
ProductId	int	Yes	Exists in Products	FK
OfferId	int	Yes	Exists in Offers	FK

SeasonalBoxes

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
ProducerId	int	Yes	Exists in Producers	FK
Name	varchar	Yes	Max 255 chars	
Description	text	No		
Season	varchar	Yes		
Price	decimal	Yes	≥ 0	
AvailableFrom	datetime	Yes	Valid date	
AvailableUntil	datetime	Yes	> AvailableFrom	

SeasonalBoxItems

Column	Data Type	Required	Validation	Key
Id	int	Yes	> 0	PK
SeasonalBoxId	int	Yes	Exists in SeasonalBoxes	FK
ProductId	int	Yes	Exists in Products	FK
Quantity	int	Yes	> 0	

AspNetUsers

Column	Type	Required	Validation	Key
Id	string	Yes		PK
UserName	string	Yes		
NormalizedUserName	string	Yes		
Email	string	Yes	Must include “@”, valid format and domain	
NormalizedEmail	string	Yes		
EmailConfirmed	bool	Yes	Must be confirmed	
PasswordHash	string	Yes		
SecurityStamp	string	No		
ConcurrencyStamp	string	No		
PhoneNumber	string	No		
PhoneNumberConfirmed	bool	No		
TwoFactorEnabled	bool	No		
LockoutEnd	datetime	No		
LockoutEnabled	bool	No		
AccessFailedCount	int	No	≥ 0	

AspNetUserRoles

Column	Type	Required	Validation	Key
UserId	string	Yes	Must exist in AspNetUsers	PK, FK
RoleId	string	Yes	Must exist in AspNetRoles	PK, FK

AspNetRoles

Column	Type	Required	Validation	Key
Id	string	Yes		PK
Name	string	Yes		
NormalizedName	string	Yes		
ConcurrencyStamp	string	No		